J. C. BOSE UNIVERSITY OF SCIENCE & TECHNOLOGY, YMCA, FARIDABAD

DEPARTMENT OF COMPUTER ENGINEERING

Progress Report for B.Tech. Capstone Project

1st Progress Report (Mar. 2026)

1. Name of Student : Vaani
2. Name of Mentor : Dr. Umesh
3. Roll No. : 24001003136
4. Name and Roll No of team member : Tanishq (24001003129)
5. Title of Project : Pneumonia Disease Detection Using Machine Learning: RespiraCheck

Progress Summary

The project has made significant progress in its core machine-learning phase. Building on the Semester 3 ResNet50 baseline (74.84% accuracy), the team upgraded the pipeline by adopting two superior backbone architectures — DenseNet121 and EfficientNetB3 — trained with a rigorous two-phase transfer learning strategy. The dataset was re-split with stratification and augmented to handle the 1:2.7 class imbalance using computed class weights. DenseNet121 was automatically selected as the best model (AUC-ROC 0.9549, Accuracy 88.78%, Pneumonia Recall 92.31%) and is now ready for backend integration. Grad-CAM++ explainability heatmaps have also been implemented and validated.

Tasks Completed

- Performed thorough **Exploratory Data Analysis (EDA)** including image sample inspection, class-count analysis, pixel-intensity distribution plots, and confirmation of the 1:2.7 NORMAL-to-PNEUMONIA class imbalance. Applied **data augmentation**: random rotations (±20°), horizontal flips, zoom (10%), and pixel shifts (10%). Computed class weights via scikit-learn to penalise the majority class during training.
- Designed a **unified classification head** appended to both backbones — GlobalAveragePooling2D → Dense(512, ReLU) → Dropout(0.5) → Dense(1, Sigmoid) — ensuring any performance difference is attributable solely to the backbone architecture.
- Executed **two-phase transfer learning** for both DenseNet121 and EfficientNetB3. Phase 1: frozen backbone, head trained for up to 10 epochs (Adam, LR = 1e-4). Phase 2: last 30 layers unfrozen (BatchNorm kept frozen), fine-tuned at LR = 1e-5. Callbacks: EarlyStopping (patience=5), ModelCheckpoint (val_auc), ReduceLROnPlateau (factor=0.5, patience=3).
- Evaluated both models on the held-out test set. **DenseNet121 was automatically selected** as the winner — Accuracy 88.78%, AUC-ROC 0.9549, Pneumonia Recall 92.31%, F1 91.14%, AP 0.9716 — and its path written to best_model_path.txt for API startup.

- Implemented **Grad-CAM++ explainability** via tf-keras-vis (v0.8.7) on DenseNet121's final convolutional layer. Heatmaps are colour-overlaid (blue → red: low → high activation) on the original X-ray and returned as Base64-encoded PNGs.
- Initialised the **GitHub repository** at https://github.com/tanishqsh2k7-source/RespiraCheck with training notebooks, saved model checkpoints, and evaluation scripts.

Tasks in Progress

- Developing the **Flask REST API** (/predict, /chat, /health endpoints) to serve the trained DenseNet121 model and Grad-CAM++ pipeline.
- Building the **React frontend** for chest X-ray image upload, prediction display, and heatmap visualisation.
- Integrating the **Google Gemini 2.5 Flash AI chat assistant** to enable patients to ask questions about their diagnosis result in accessible language.
- Planning end-to-end **integration testing** of the full pipeline — image upload → model inference → Grad-CAM++ → JSON response → frontend render.

Date: ……………… **(Signature of Student)**

Report of the Mentor

(to be filled by mentor)

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Signature of Mentor